

AILT - Study 2 (#127197)

Author(s)

Chiara Longoni (Boston University) - chiara.longoni2@unibocconi.it
Stephanie Tully (University of Southern California) - tullyste@marshall.usc.edu
Gil Appel (George Washington University) - gappel@gwu.edu

Pre-registered on: 2023/03/30 - 11:25 AM (PT)

1) Have any data been collected for this study already?

No, no data have been collected for this study yet.

2) What's the main question being asked or hypothesis being tested in this study?

We are examining the relationship between algorithmic knowledge and propensity to use AI tools to complete assignments.

3) Describe the key dependent variable(s) specifying how they will be measured.

The dependent measure will be propensity to use an AI tools to complete 4 different assignments on the following scale: 1 - I would not use the generative AI agent, 2 - I would use the generative AI agent to come up with ideas, but would write the assignment myself, 3 - I would use the generative AI agent to write the first draft, but would re-write it before submitting it, 4 - I would use the generative AI agent to complete the assignment, but would fact check it to fix any errors before submitting it, 5 - I would use the generative AI agent to complete the assignment, and submit it as written

4) How many and which conditions will participants be assigned to?

There will be no conditions. Our independent variable will be participants' algorithmic literacy as measured using a 25-item multiple choice test. The sum of the correct responses will serve as their knowledge level.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

We will calculate participants' propensity to use AI across 4 different assignments. We will then regress this average use propensity on their algorithmic literacy score

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

We include one instructional manipulation check (IMC; "This question is NOT interested in your marital status. Instead, it is designed to ensure you are taking the time to read all questions, even towards the end of the survey. As such, instead of indicating your marital status from the options below, please select "married" from the options below."). We will conduct our analyses excluding those who failed the IMC. There will be no conditions. Our independent variable will be participants' algorithmic literacy as measured using a 25-item multiple choice test. The sum of the correct responses will serve as their knowledge level.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

We will run this survey in four weekly lab sessions. Each week could have up to a maximum of 100 people. However, the final sample will depend on the number of people who participate in the lab sessions over this 4 week time period.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

We will collect demographic information